



VIT

Vellore Institute of Technology

Continuous Assessment Test II - March 2025

Programme	B. Tech CSE and its specialization	Semester	WIN 2024 - 25
Course Code	BCSE324L	Slot	G2+TG2
Course Title	Foundations of Blockchain Technology	Class Number	CH2024250502197
Faculty(s)	Dr. R. Prabhakaran	Max. Marks	50
Time	90 Minutes		

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information.

Q No.	Sub-Sec	Description	Marks
1.		VIT University is considering adopting a Decentralized Autonomous Organization (DAO) platform, such as Colony DAO, to manage student-led projects, research initiatives, and funding allocations. The university aims to enhance transparency, efficiency, and student participation in decision-making processes. However, there are concerns about the technical complexity, security risks, and the readiness of students and faculty to adapt to such a decentralized system.	
	a)	Analyze the potential benefits and challenges of implementing Colony DAO at VIT University for managing student projects and funding. (5 marks)	10
	b)	Evaluate the feasibility of adopting Colony DAO at VIT University and necessary steps should be taken to address concerns related to technical complexity, security, and user adoption. (5 marks)	
		SRS Infotech company is exploring the use of smart contracts to automate client agreements, payment terms, and project milestones. The company aims to improve transparency, reduce administrative overhead, and ensure timely execution of contractual obligations. Conversely, there are concerns about the technical complexity, legal enforceability, and the need for employee training to adopt this technology.	
	a)	Discuss in detail how SRS Infotech company ensures the legal enforceability and seamless integration of smart contracts while addressing the technical complexities and upskilling employees for effective adoption. (5 marks)	
	b)	Illustrate how might the use of state variables, functions, and gas fees impact the company's operational efficiency and client relationships. (5 marks)	
		ABC Digital Marketing Pvt Ltd. Company is considering adopting blockchain-based Decentralized Autonomous Organizations (DAOs) to enhance transparency, streamline collaboration, and manage its marketing campaigns effectively. The company is particularly interested in using DAOs for decision-making, budget management, and project approvals.	
	a)	Describe in detail how implementing DAOs could improve the transparency and accountability of decision-making processes within the company. (5 marks)	

	b)	Evaluate the potential challenges ABC Digital Marketing may encounter when using DAOs for managing marketing projects, and suggest strategies to mitigate these challenges. (5 marks)
4.		A state government is exploring the implementation of blockchain-based governance systems to enhance transparency, streamline decision-making, and improve public service delivery. The government aims to leverage blockchain for secure voting, transparent budgeting, and efficient management of public resources. (5 marks)
	a)	Examine how blockchain governance models, including on-chain and off-chain governance, can be applied to enhance transparency and accountability in state government operations.
	b)	Assess the challenges a state government might face in implementing blockchain-based governance systems and suggest practical solutions to address these challenges. (5 marks)
		MGM Air Transport is exploring the adoption of smart contracts to enhance operational efficiency in cargo management, fleet tracking, and insurance claims processing. By leveraging blockchain technology, the company aims to ensure transparency, automate processes, and reduce administrative overhead.
	a)	Explore how the stages of the smart contract life cycle can be applied to automate and streamline MGM Air Transport's supply chain and cargo management operations. (5 marks)
	b)	Apply the smart contract pattern for MGM Air Transport to automate the transport operation effectively. (5 marks)